

Bri's Battery Documentation

7/6/2026-8/13/2026

State Machine Prototype (C++)

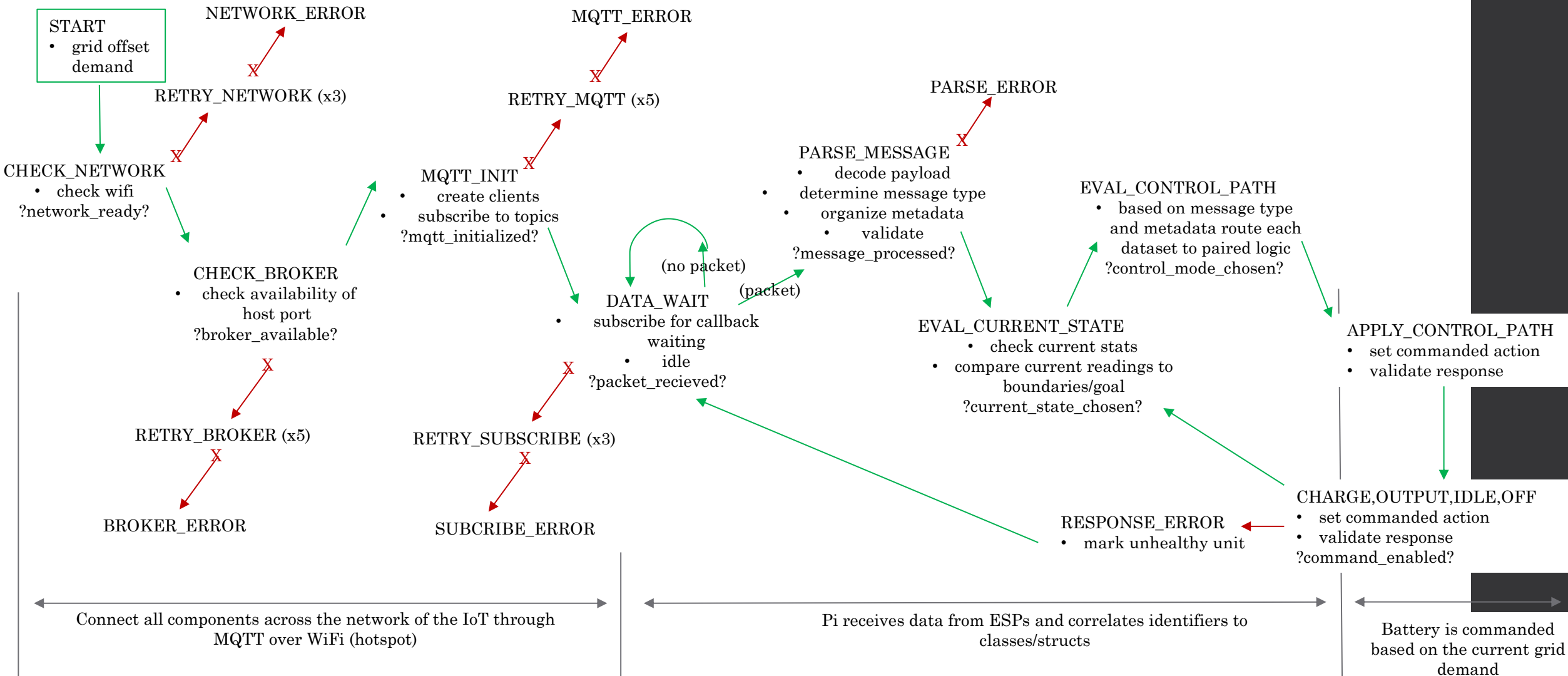
Key:

CAPS = state

- = action

→ /**X** = transition (true/false branch)

?<boolean>? = condition

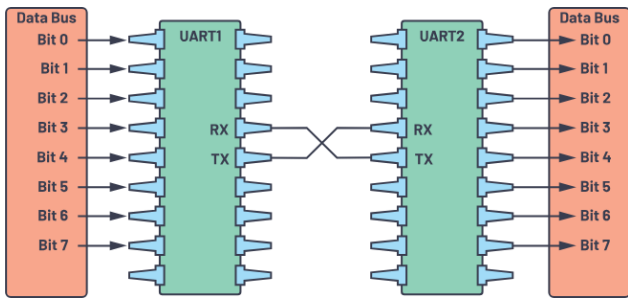


Class vs Struct Data

- AC/DC Output Commands
- Plug/Relay On/Off Commands
- Telemetry Reads
- Log Data
- Publish MQTT
- Connect
- Disconnect
- Poll Scheduling/Timing
- Checks
- Detect/Categorize Faults
- Retry Failed Connection
- User Human Readable Alerts
- Save/load Configuration

- Total Battery Percentage (SoC)
- AC/DC Output On
- AC/DC Input Power
- AC/DC Output Power
- Power Generation
- Plug Voltage (V)
- Plug Current (A)
- Plug Power (W)
- Plug Energy (kWh)
- Plug Frequency (Hz)
- MQTT Status
- Uptime
- Firmware Version
- Wifi-RSSI
- IP Address
- Serial
- Dsp Version
- Arm Version
- Error Flags
- Alarm Flags
- ESP32 Uptime
- ESP32 Temperature

Troubleshooting the ESP to Pi MQTT Bridge Using UART



Command	Result
<pre>Serial.begin(<baud>); void loop(){ Serial.println("<string>"); }</pre>	Establishes shared connection transmission speed and tells the UART peripheral to transmit data bytes

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

* Executing task: C:\Users\HP\.platformio\p

o\penv\Scripts\platformio.exe device monit

or

(TX)

LVGL Setup done

deactivate relais contact

Loading Values from EEPROM

*wm:AutoConnect

*wm:Connecting to SAVED AP: hyperview-ap

*wm:connectTimeout not set, ESP waitForCon

nectResult...

*wm:AutoConnect: FAILED

*wm:StartAP with SSID: Bluetti_ESP32

*wm:AP IP address: 192.168.4.1

Entered config mode:ip=192.168.4.1, ssid='

Bluetti_ESP32'

*wm:Starting Web Portal

hn":"plug-6177","mac":"C4D8D50CF821","md":

"Sonoff Basic","ty":0,"if":0,"cam":0,"ofln

":"Offline","onln":"Online","state":["OFF"

,"ON","TOGGLE","HOLD"],"sw":"15.3.0.3","t"

:"plug","ft":"%prefix%/%topic%/", "tp":["cm

nd","stat","tele"],"rl":[1,0,0,0,0,0,0,0,0

,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0

,0,0],"swc":[-1,-1,-1,-1,-1,-1,-1,-1,-1,-1

, -1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1

, -1,-1,-1,-1,-1,-1,-1,-1],"swn":[null,null

,null,null,null,null,null,null,null,null,n

ull,null,null,null,null,null,null,null,nul

l,null,null,null,null,null,null,null,null,

null,null,null,null,null],"btn":[0,0,0,0,

0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,

0,0,0,0,0,0,0],"so":{"4":0,"11":0,"13":0,"1

7":0,"20":0,"30":0,"68":0,"73":0,"82":0,"1

14":0,"117":0},"lk":0,"lt_st":0,"bat":0,"d

brianna@hyperview:~ \$ sudo journalctl -f

Apr 11 21:58:18 hyperview sshd-session[9149

] : Accepted password for brianna from 10.42

.0.44 port 51708 ssh2

Apr 11 21:58:18 hyperview sshd-session[9149

] : pam_unix(sshd:session): session opened f

or user brianna(uid=1000) by brianna(uid=0)

Apr 11 21:58:18 hyperview systemd-logind[66

1] : New session 6 of user brianna.

Apr 11 21:58:18 hyperview systemd[1] : Start

ed session-6.scope - Session 6 of User bria

nna.

Apr 11 21:58:39 hyperview sshd-session[9393

] : Accepted password for brianna from 10.42

.0.44 port 49235 ssh2

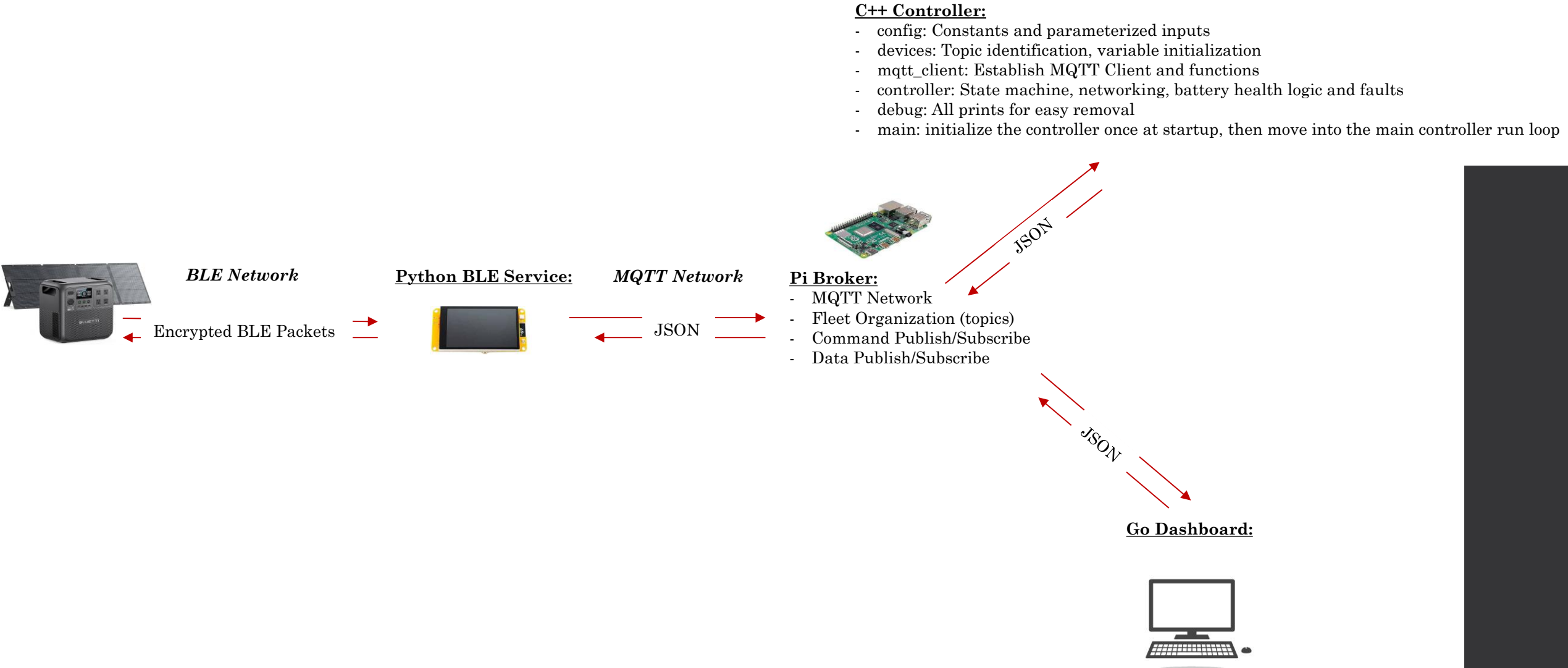
Apr 11 21:58:39 hyperview sshd-session[9393

] : pam_unix(sshd:session): session opened f

or user brianna(uid=1000) by brianna(uid=0)

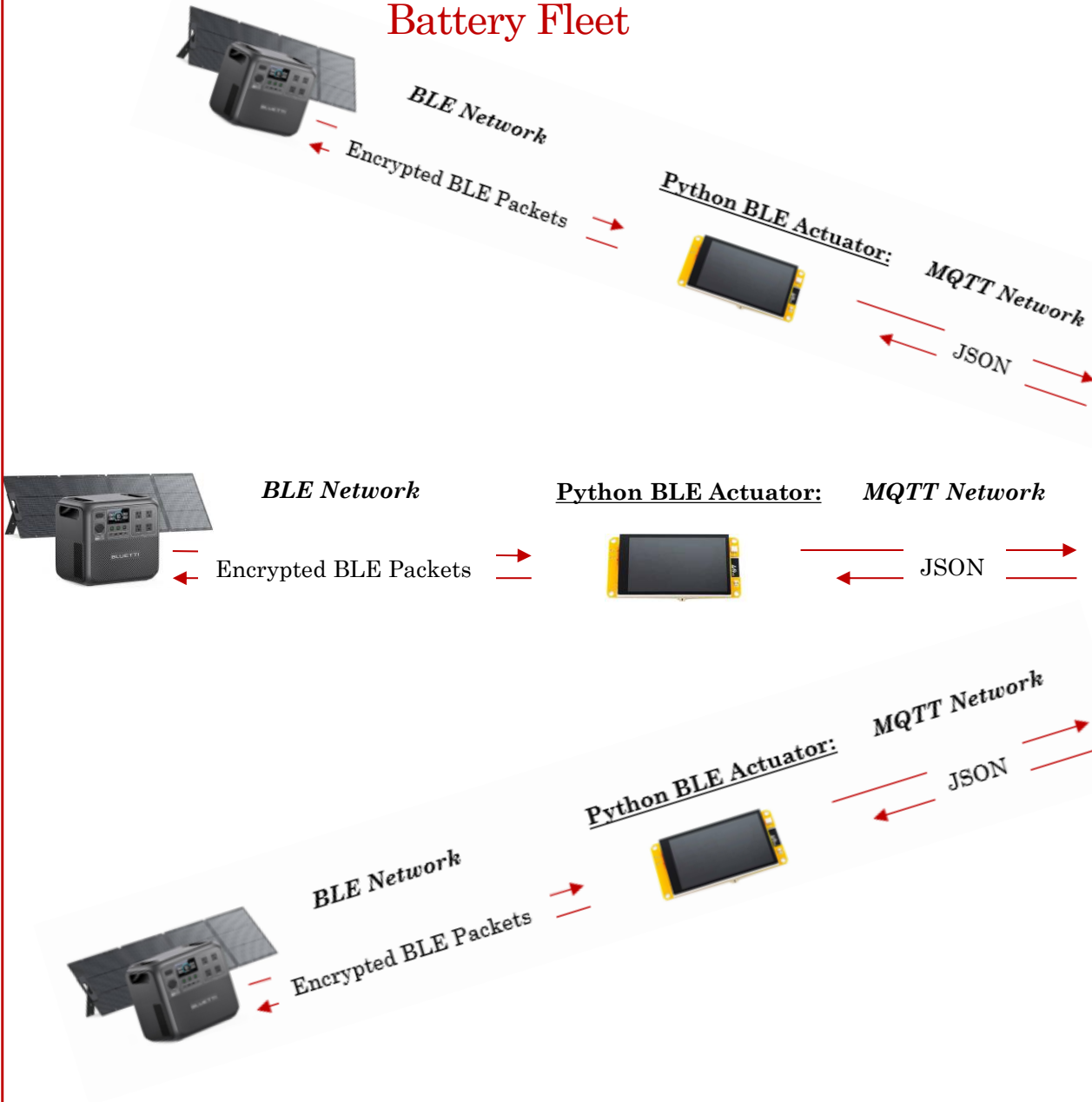
Ln 351, Col 1 (11266 selected) Spaces: 2

System Inter-Language Coms Prototype:



System Prototype:

Battery Fleet



Local Broker



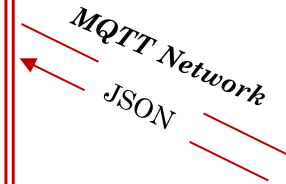
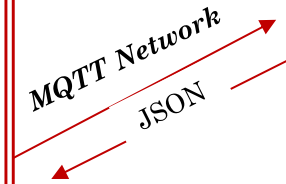
Pi Broker:

- Fleet organization (topics)
- Command publish/subscribe
- Data and telemetry publish/subscribe

Remote Management

C++ Controller:

- Tracks data from battery fleet and evaluates health/faults
- Takes input grid demand, determines schedule
- Sends command packet back to actuator



Go Dashboard:

- Displays data
- Tracks trends across time



JSON Parsing (Python, C++, and Go)

docs.python.org/3/library/json.html

3.14.6 Quick search Go

json — JSON encoder and decoder

Source code: [Lib/json/_init_.py](#)

github.com/nlohmann/json

nlohmann / json

Code Issues 29 Pull requests 58 Agents Discussions Actions Projects More

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JSON for Modern C++

MIT License

[json.nlohmann.me](#)

pkg.go.dev/encoding/json#example-Decoder

GO json package standard library

Version: go1.26.5 Latest | Published: Jul 7, 2026 | Licenses

Jump to ...

Documentation

- Overview
- Index
- Examples
- Constants

Types

type Decoder

```
type Decoder struct {  
    // contains filtered or unexported fields  
}
```

A Decoder reads and decodes JSON values from an input stream.

```
23 def json_publish_battery_state(state):  
24     payload = {  
25         "soc": state.soc,  
26         "ac_output": state.ac_output,  
27         "dc_output": state.dc_output,  
28     }  
29     mqtt_client.publish(  
30         config.TOPIC_TELEMETRY,  
31         json.dumps(payload)  
32     )  
33  
34 def json_process_message(msg):  
35     payload = json.loads(msg.payload)  
36     return {  
37         "soc": payload.get("soc"),  
38         "ac_output": payload.get("ac_output"),  
39         "charge_limit": payload.get("charge_limit"),  
40     }
```

```
static DeviceInformation latestBattery; // temporary storage for the newest telemetry received  
static void OnMessage(  
    mosquitto*,  
    void*,  
    const mosquitto_message* msg){  
  
    //display received topic for debugging  
    std::cout<<"Topic: "<<msg->topic<<std::endl;  
  
    //display received payload for debugging  
    std::cout<<"Payload: "<<std::string(static_cast<char*>(msg->payload),msg->payloadlen)<<std::endl;  
    //later:  
    //assemble payload  
    std::string payload(static_cast<char*>(msg->payload), msg->payloadlen);  
    try{  
        json data = json::parse(payload);  
        std::cout << std::setw(4) << data << "\n\n"; //serialize json  
  
        //update storage container variables with new data  
        latestBattery.SOC = data["soc"];  
        latestBattery.batteryCurrent = data["current"];  
    }  
    catch(const json::exception& e){  
        std::cerr << e.what() << std::endl;  
    }  
}
```


Successful Automated Package Delivery From Python to C++ Through JSON!

```
[READ] DC Input Switch: [0]
[Read AC Input Switch] 00 08 b9 e8 d3 0c e9 00 aa 3
3 2e 6f 9b a5 c7 b0 5b 30 fc 3b b3 83
Finished storing values inside variables
[READ] AC Input Switch: [0]
[Read Grid Charging Permission] 00 08 cc 05 6d bf b
9 47 c4 7d d4 10 16 0b c3 3b 19 d1 6e ch 25 c9
Finished storing values inside variables
[READ] Grid Charging Permission: [1]
[Read AC Input Setting] 00 08 42 50 a1 ad 0a 2c d4
1b 30 fc d5 a4 31 d9 74 98 9e 0c fc eb
Finished storing values inside variables
[READ] AC Input Setting: [0]
[Read Home Telemetry Block] 00 08 96 78 50 fe 68 76
dd c9 26 09 bb 25 0a 0a 0d f5 69 3d 27 f6
Finished storing values inside variables
[READ] Home Telemetry Block: [3986, 28, 97, 2, 1986
, 1986, 0, 1, 1, 0, 19525, 12340, 48, 0, 0, 0, 3470
0, 22612, 606, 0, 1, 0, 1, 648, 2578, 0, 0, 0, 0, 0
, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 84, 0, 0]
[Read AC/DC Load Info Block] 00 08 21 4a 94 ed 07 8
b ef 86 a7 e8 44 54 8b 4f 1e 1a 01 a9 f7 10
Finished storing values inside variables
[READ] AC/DC Load Info Block: [0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 84, 0, 0, 0
, 0, 0, 0, 0, 0, 1, 83, 1196, 9, 0, 0]
[Read PV Input Info Block] 00 08 4b 6c 6b 9b f8 2d
6e 84 70 be bf d1 a1 a0 12 51 d7 6d 90 bb
Finished storing values inside variables
[READ] PV Input Info Block: [0, 0, 95, 0, 0, 0, 0,
0, 0, 0, 1]
[]
```

```
System has a fault
System has a fault
PS C:\ecolibrum\smart_plug_PI> .\BatteryController.exe
STARTING CONTROLLER
Connected to MQTT broker.
Subscribed to tele/battery/STATE
System is idle
System has a fault
System has a fault
System has a fault
System has a fault
Topic: tele/battery/STATE
Payload: {"device": "bluetti_ac200", "timestamp": 178596284
1.0000385, "action": {"request": "read", "target": null, "v
alue": null, "status": null, "state": "idle"}, "telemetry":
{"home_telemetry": {"soc": 97, "voltage": 399.2, "current"
: 0.1, "pv_watts": 0, "grid_watts": 0, "ac_watts": 0, "dc_w
atts": 0}, "dc_output": {"value": 0}, "ac_output": {"value"
: 1}, "dc_input": {"value": 0}, "ac_input": {"value": 0}, "
grid_permission": {"value": 1}, "ac_input_setting": {"value
": 0}}}
timestamp
JSON parse failed: [json.exception.type_error.302] type mus
t be string, but is null
Failed to parse battery telemetry.
System has a fault
System has a fault
System has a fault
System has a fault
System has a fault
System has a fault
PS C:\ecolibrum\smart_plug_PI> █
```

```
"[[{"tele/battery/STATE {"device": "bluetti_ac2
00", "timestamp": 1785962097.434542, "action": {"r
equest": "read", "target": null, "value": null, "s
tatus": null, "state": "idle"}, "telemetry": {"hom
e_telemetry": {"soc": 97, "voltage": 399.2, "curre
nt": 0.1, "pv_watts": 0, "grid_watts": 0, "ac_watt
s": 0, "dc_watts": 0}, "dc_output": {"value": 0},
"ac_output": {"value": 1}, "dc_input": {"value": 0
}, "ac_input": {"value": 0}, "grid_permission": {"
value": 1}, "ac_input_setting": {"value": 0}}}
tele/battery/STATE {"device": "bluetti_ac200", "ti
mestamp": 1785962111.246163, "action": {"request":
"read", "target": null, "value": null, "status":
null, "state": "idle"}, "telemetry": {"home telem
etry": {"soc": 97, "voltage": 399.2, "current": 0.0
, "pv_watts": 0, "grid_watts": 0, "ac_watts": 0, "
dc_watts": 0}, "dc_output": {"value": 0}, "ac outp
ut": {"value": 1}, "dc_input": {"value": 0}, "ac_i
nput": {"value": 0}, "grid_permission": {"value":
1}, "ac_input_setting": {"value": 0}}}
tele/battery/STATE {"device": "bluetti_ac200", "ti
mestamp": 1785962125.1193569, "action": {"request"
: "read", "target": null, "value": null, "status":
null, "state": "idle"}, "telemetry": {"home telem
etry": {"soc": 97, "voltage": 399.2, "current": 0.
1, "pv_watts": 0, "grid_watts": 0, "ac_watts": 0,
"dc_watts": 0}, "dc_output": {"value": 0}, "ac_out
put": {"value": 1}, "dc_input": {"value": 0}, "ac
input": {"value": 0}, "grid_permission": {"value":
1}, "ac_input_setting": {"value": 0}}}
tele/battery/STATE {"device": "bluetti_ac200", "ti
mestamp": 1785962139.0119865, "action": {"request"
```